DILLIES'S CONVEX VC_2 CONJECTURE IN THREE DIMENSIONS

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Abstract. We prove that every convex subset of $\mathbb{R}^3$ has additive VC_2 dimension at most two, resolving the corresponding conjecture in the Formal Conjectures library. Equivalently, no additive 3×3 array can be shattered by translations of one convex set. The proof identifies a two-pattern obstruction. The nine patterns omitting one array entry force a canonical form for the two summand triangles. Two further patterns, each omitting two entries, supply four strictly separating functionals. After positive normalization, these functionals occupy the bottom, top, right, and left strips surrounding a unit square. The translation difference would have to evaluate positively on the first two and negatively on the last two, which is impossible for an affine function. The argument uses separation only from finite convex hulls and therefore applies to arbitrary convex sets, including open and unbounded sets. A single Lean 4 source file proves the repository's original statement; its axiom audit contains only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`.

1. Introduction

The VC dimension of a family of translates measures its ability to realize arbitrary membership patterns on finite sets. Convexity imposes a strong restriction in the plane, but a different picture emerges in higher dimensions. Naszódi and Taschuk [6] proved that the VC dimension of families of positive homothets of a planar convex body is at most three, whereas some convex bodies in higher dimensions have families of translates of infinite VC dimension.

Additive VC_n dimension restricts the sets to be shattered to arrays formed by sums of n independently indexed sequences. For $n = 2$, the points form an additive grid. The Formal Conjectures library [2] contains a conjecture proposed by Yaël Dillies [1]: every convex set in $\mathbb{R}^3$ has additive VC_2 dimension at most two. We prove this assertion.

Definition 1.1. A set C in a real vector space has additive VC_2 dimension at most d if there are no sequences $a_0, \ldots, a_d$ and $b_0, \ldots, b_d$ such that, for every $S \subseteq \{0, \ldots, d\}^2$, some translation t_S satisfies

$$t_S + a_i + b_j \in C \quad \Longleftrightarrow \quad (i, j) \in S \qquad (0 \le i, j \le d). \tag{1}$$

When such translations exist, we say that C *shatters* the additive array $(a_i + b_j)_{i,j}$.

Theorem 1.2. *Every convex set $C \subseteq \mathbb{R}^3$ has additive VC_2 dimension at most two.*

The bound is optimal: existing convex examples in $\mathbb{R}^3$ shatter an additive 2×2 array [3]. A general finite bound has also been formalized [4]: for $n \ge 1$, convex subsets of $\mathbb{R}^{n+1}$ have additive VC_n dimension at most $2^{8(n+2)^n} - 1$. At $n = 2$ this gives $2^{128} - 1$; Theorem 1.2 establishes the conjectured bound two.

The proof has two parts. An algebraic analysis of common exposing functionals forces an affine normal form for every potentially shattered 3×3 array. In that normal form, two specific membership patterns are incompatible. The obstruction is a planar affine-separation argument in the space of normalized functionals. In particular, the proof does not require compactness, closedness, or an interior assumption on C.

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2. Finite separation and exposed arrays

Throughout the proof, $a, b : \{0,1,2\} \to \mathbb{R}^3$ are triples of vectors. All exposing functionals are real linear functionals.

Lemma 2.1 (Finite strict separation)**.** *Let $C \subseteq \mathbb{R}^3$ be convex, let $F \subseteq C$ be finite, and let $q \notin C$. There is a linear functional f such that*

$$f(p) < f(q) \qquad (p \in F).$$

Proof. The convex hull $K = \operatorname{conv} F$ is compact and is contained in C, so $q \notin K$. Strict separation of a point from a compact convex set gives the required functional. If F is empty, the assertion is vacuous. □

Definition 2.2. The pair of triples (a, b) is an *exposed array* if, for every $i, j \in \{0,1,2\}$, there is a linear functional f satisfying

$$f(a_k) < f(a_i) \quad (k \neq i), \qquad f(b_\ell) < f(b_j) \quad (\ell \neq j).$$

Lemma 2.3. *If a convex set C shatters $(a_i + b_j)_{i,j\in\{0,1,2\}}$, then (a, b) is an exposed array. Moreover, the nine sums are distinct vertices of their convex hull.*

Proof. For a fixed (i, j), choose the membership pattern containing the other eight entries and omitting this one. If the omitted sum belonged to the convex hull of the other sums, the corresponding translated point would belong to C by convexity, a contradiction. Thus each sum is outside the convex hull of the others.

Alternatively, apply Lemma 2.1 to the eight included translated points and the excluded point. Comparing within the same column cancels the translation and the b-summand, and comparing within the same row cancels the translation and the a-summand. This gives Definition 2.2. Equal sums could not receive different membership values, so the nine sums are distinct. □

3. An algebraic normal form

We use affine relations with separately vanishing coefficient sums:

$$\sum_{k=0}^{2} \alpha_k = 0, \qquad \sum_{\ell=0}^{2} \beta_\ell = 0, \qquad \sum_{k=0}^{2} \alpha_k a_k + \sum_{\ell=0}^{2} \beta_\ell b_\ell = 0. \tag{2}$$

Lemma 3.1 (Common-sign relations)**.** *Suppose (a, b) is an exposed array and (2) holds. If, for some $i, j \in \{0,1,2\}$,*

$$\alpha_k \leq 0 \quad (k \neq i), \qquad \beta_\ell \leq 0 \quad (\ell \neq j),$$

then all six coefficients are zero. The same conclusion holds when both inequalities are reversed.

Proof. Choose a functional f exposing the pair (a_i, b_j). The two vanishing coefficient sums and the vector relation give

$$0 = \sum_k \alpha_k \big(f(a_k) - f(a_i)\big) + \sum_\ell \beta_\ell \big(f(b_\ell) - f(b_j)\big).$$

Every summand is nonnegative. For $k \neq i$, the second factor in the first sum is strictly negative, so its summand vanishes only if $\alpha_k = 0$. Similarly $\beta_\ell = 0$ for $\ell \neq j$. The vanishing coefficient sums then imply $\alpha_i = \beta_j = 0$. Negating the relation proves the assertion for reversed inequalities. □

Lemma 3.2 (A zero coefficient forces triviality)**.** *For an exposed array, any relation (2) with a zero coefficient is trivial.*

Proof. Suppose $\alpha_i = 0$. The other two α-coefficients sum to zero. Consequently there is an index outside which all α-coefficients are nonpositive, and an index outside which they are all nonnegative. This includes the case when all three are zero.

Among any three real numbers, at least two are nonnegative or at least two are nonpositive. Apply this observation to the β-coefficients, and choose j to be the remaining index. Choose the α-index whose

inequality direction matches that of the two β-coefficients. Lemma 3.1 makes the whole relation trivial. The case $\beta_j = 0$ follows by interchanging the two triples. □

Lemma 3.3 (A forced basis)**.** *If (a, b) is an exposed array, the vectors*

$$a_1 - a_0, \qquad a_2 - a_0, \qquad b_1 - b_0$$

are linearly independent. In $\mathbb{R}^3$ they therefore form a basis.

Proof. A relation $s(a_1 - a_0) + t(a_2 - a_0) + r(b_1 - b_0) = 0$ has the form (2) with

$$\alpha = (-s - t, s, t), \qquad \beta = (-r, r, 0).$$

The zero coefficient β_2 and Lemma 3.2 imply $s = t = r = 0$. □

Proposition 3.4 (Canonical normal form)**.** *If (a, b) is an exposed array in $\mathbb{R}^3$, then after separately permuting the two triples and making an invertible affine change of coordinates, its sums have the form*

$$A = (0, e_1, e_2), \qquad B = (0, e_3, (u, v, w)), \qquad u < 0,\ v < 0,\ 0 < w < 1. \tag{3}$$

Proof. The four vectors $a_1 - a_0, a_2 - a_0, b_1 - b_0, b_2 - b_0$ are linearly dependent in $\mathbb{R}^3$. Expanding a nontrivial relation between them gives (2) with coefficients not all zero. Lemma 3.2 implies that every coefficient is nonzero. Each coefficient triple sums to zero, so one sign occurs once and the opposite sign occurs twice.

The two solitary signs must be opposite. If they agreed, choosing the two solitary indices and, if necessary, negating the relation would give precisely the hypotheses of Lemma 3.1. That would make the relation trivial.

Relabel and orient the relation so that

$$\alpha_0 < 0, \quad \alpha_1, \alpha_2 > 0, \qquad \beta_0, \beta_1 < 0, \quad \beta_2 > 0.$$

Set

$$u = -\frac{\alpha_1}{\beta_2}, \qquad v = -\frac{\alpha_2}{\beta_2}, \qquad w = -\frac{\beta_1}{\beta_2}.$$

Then $u, v < 0$ and $w > 0$. Also $\beta_0 + \beta_1 + \beta_2 = 0$ and $\beta_0 < 0$ imply $w < 1$. Using the two coefficient sums in the vector relation gives

$$b_2 - b_0 = u(a_1 - a_0) + v(a_2 - a_0) + w(b_1 - b_0). \tag{4}$$

By Lemma 3.3, the three directions on the right form a basis. Subtract a_0, b_0 from their respective triples and carry this basis to e_1, e_2, e_3 by an invertible linear map. This gives (3). Translating the origin and applying an invertible linear map preserve convexity and all translation membership patterns. □

Remark 3.5. No general-position hypothesis appears in Proposition 3.4. The algebraic argument excludes repeated, collinear, coplanar, and other degenerate configurations whenever they obstruct the required common exposures.

4. The two-pattern obstruction

Work in the normal form (3), and index the sums by $X_{3i+j} = a_i + b_j$. Thus

$$\begin{array}{lll} X_0 = 0, & X_1 = e_3, & X_2 = (u, v, w), \\ X_3 = e_1, & X_4 = e_1 + e_3, & X_5 = e_1 + (u, v, w), \\ X_6 = e_2, & X_7 = e_2 + e_3, & X_8 = e_2 + (u, v, w). \end{array} \tag{5}$$

Consider the two included index sets

$$S = \{0, \dots, 8\} \setminus \{0, 1\}, \qquad T = \{0, \dots, 8\} \setminus \{5, 8\}. \tag{6}$$

Their membership matrices, with 1 denoting inclusion, are

$$S : \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad T : \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

Lemma 4.1 (Four-strip obstruction)**.** *Let N_0, N_1, N_5, N_8 be points in the following four open strips:*

	ξ	η
N_0	$0 < \xi < 1$	$\eta < 0$
N_1	$0 < \xi < 1$	$\eta > 1$
N_5	$\xi > 1$	$0 < \eta < 1$
N_8	$\xi < 0$	$0 < \eta < 1$.

No affine function $\ell(\xi, \eta) = \alpha\xi + \beta\eta + \gamma$ is positive at N_0, N_1 and negative at N_5, N_8.

Proof. Coordinatewise ordering gives the following four contradictions:

coefficient signs	forced inequality
$\alpha \geq 0,\ \beta \geq 0$	$\ell(N_5) \geq \ell(N_0) > 0$
$\alpha \geq 0,\ \beta \leq 0$	$\ell(N_5) \geq \ell(N_1) > 0$
$\alpha \leq 0,\ \beta \geq 0$	$\ell(N_8) \geq \ell(N_0) > 0$
$\alpha \leq 0,\ \beta \leq 0$	$\ell(N_8) \geq \ell(N_1) > 0$.

These cases cover zero coefficients as well. Geometrically, the segments joining the bottom and top points and the left and right points cross inside the square; they cannot lie in opposite strict affine halfplanes. □

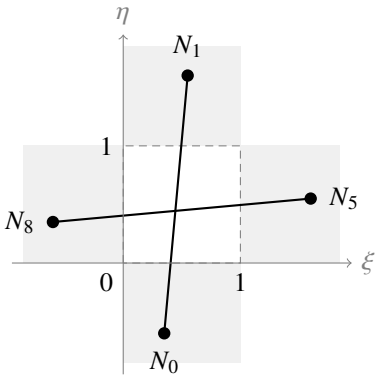


FIGURE 1. The four strips in the normalized normal plane. The locations shown are illustrative; the obstruction uses only the strip inequalities.

Proposition 4.2. *For an array in the normal form (3), no convex $C \subseteq \mathbb{R}^3$ can realize both patterns in (6) by translations.*

Proof. Suppose t, z realize S, T, respectively. The convex hull

$$K = \operatorname{conv}\big(\{X_i + t : i \in S\} \cup \{X_j + z : j \in T\}\big)$$

is contained in C. The four excluded points $X_0 + t, X_1 + t, X_5 + z, X_8 + z$ lie outside K. Choose strict separating normals n_0, n_1, n_5, n_8, with each normal taking a larger value at its excluded point than at every point of K. They exist by Lemma 2.1.

For a normal $n = (p, q, r)$ put $c = up + vq$. Comparisons with included points at the same translation give

$$\begin{array}{c|l} n_0 & p < 0,\ q < 0,\ c + wr < 0 \\ n_1 & p < 0,\ q < 0,\ c - (1-w)r < 0 \\ n_5 & p > 0,\ c + wr > 0,\ c - (1-w)r > 0 \\ n_8 & q > 0,\ c + wr > 0,\ c - (1-w)r > 0. \end{array} \tag{7}$$

For the four rows compare, respectively, with grid indices $(3, 6, 2)$, $(4, 7, 2)$, $(2, 3, 4)$, and $(2, 6, 7)$. For n_0, n_1, the signs of u, v, p, q imply $c > 0$. For n_5, n_8, use

$$(1-w)(c + wr) + w\big(c - (1-w)r\big) = c > 0.$$

Divide each normal by its positive c. All four now satisfy $up + vq = 1$. On this affine plane introduce coordinates

$$\xi = vq = 1 - up, \qquad \eta = w(1-w)r + 1 - w. \tag{8}$$

The first normal has $0 < \xi < 1$ and $\eta < 0$; the second has $0 < \xi < 1$ and $\eta > 1$. For the third, $p > 0$ gives $\xi > 1$, and the two bounds on r give $0 < \eta < 1$. For the fourth, $q > 0$ gives $\xi < 0$, with the same bounds on η. Thus their coordinate points occupy precisely the four strips of Lemma 4.1.

Set $d = t - z$. The other pattern includes each excluded grid cell: $X_0 + z, X_1 + z, X_5 + t, X_8 + t \in K$. Strict separation at the four excluded points therefore implies

$$n_0 \cdot d > 0, \qquad n_1 \cdot d > 0, \qquad n_5 \cdot d < 0, \qquad n_8 \cdot d < 0. \tag{9}$$

Positive normalization preserves these signs. Evaluation at d is an affine function on the normal plane. Explicitly, the inverse of (8) is

$$p = \frac{1-\xi}{u}, \qquad q = \frac{\xi}{v}, \qquad r = \frac{\eta - 1 + w}{w(1-w)}.$$

Writing $d = (d_1, d_2, d_3)$ gives

$$n \cdot d = \left(\frac{d_2}{v} - \frac{d_1}{u}\right)\xi + \frac{d_3}{w(1-w)}\eta + \frac{d_1}{u} - \frac{d_3}{w}.$$

The signs (9) contradict Lemma 4.1. □

Proof of Theorem 1.2. Suppose a convex $C \subseteq \mathbb{R}^3$ shatters an additive 3×3 array. Lemma 2.3 makes it an exposed array. Proposition 3.4 puts it in the canonical normal form, preserving all membership patterns. Full shattering requires both patterns in (6), contrary to Proposition 4.2. □

5. Formalization and verification

The complete proof is contained in the single file `DilliesConvexVC2Conjecture.lean`. It imports the additive VC-dimension definition and the three-dimensional Euclidean-space notation directly from Formal Conjectures. The final declaration has the original name

`VCDimConvex.hasAddVCNDimAtMost_two_two_of_convex_r3`

and the original type

$$\forall\{C : \mathrm{Set}(\mathbb{R}^3)\}, \quad \mathrm{Convex}_{\mathbb{R}}(C) \longrightarrow \texttt{HasAddVCNDimAtMost}\ C\ 2\ 2.$$

In particular, the theorem adds no closedness, boundedness, dimension, interior, or nondegeneracy hypothesis.

The formal proof follows the algebraic reduction in Section 3. For the final contradiction it evaluates each separating functional on the basis directions $a_1 - a_0, a_2 - a_0, b_1 - b_0$. This produces the same coordinates without explicitly changing the ambient convex set. The translation difference is expanded in that basis. Finite strict separation uses Mathlib's [5] `geometric_hahn_banach_closed_point` after replacing the included points by their compact convex hull.

Mathematical step	Lean declaration in `ConvexVC2Dimension`
Finite strict separation	`finite_separator`
Shattering forces exposure	`GridShattered.exposed`
Zero coefficients force triviality	`relation_zero_of_zero_entry`
Forced basis	`exposed_three_independent`
Canonical normal-form relation	`exposed_normal_form`
Four-strip obstruction	`cross_strip_impossible`
Positive normalization	`raw_normal_obstruction`
Translation contradiction	`canonical_grid_not_shattered`

Verification used Lean 4.33.1, Mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`, and Formal Conjectures revision `8432eac998110a563e03df65a28c117e97c8c142`. The file compiles with warnings treated as errors. The final `##print axioms` command reports exactly

[`propext`, `Classical.choice`, `Quot.sound`].

There are no admitted proofs, custom axioms, or calls to `native_decide` in the file. Exploratory computations helped locate the two patterns, but no numerical or external solver result is a premise of the formal proof.

6. The finite obstruction

The proof uses only eleven required patterns after the appropriate relabeling: the nine patterns omitting one cell force exposure and the normal form, and the two patterns in (6) give the contradiction. The incompatible two-pattern family is therefore identified geometrically rather than found by exhaustively testing all 512 labels.

The shared convex set is essential in the separating-functional step: each normal must separate its excluded point from included points at *both* translations. Comparisons at the same grid cell then turn the translation difference into one affine function on the normalized normal plane. The crossing obstruction applies to this single function.

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